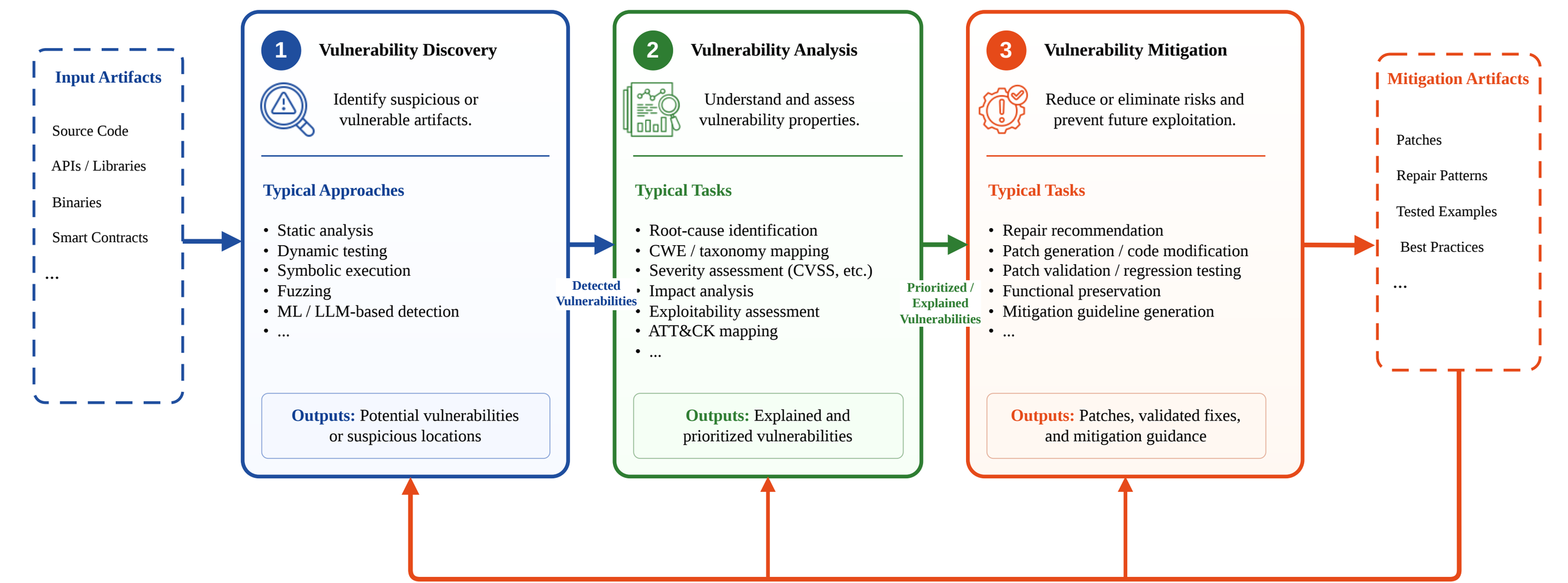

Input Artifacts
Source Code
APIs / Libraries
Binaries
Smart Contracts
...
1
Vulnerability Discovery
Identify suspicious or vulnerable artifacts.
Typical Approaches
• Static analysis
• Dynamic testing
• Symbolic execution
• Fuzzing
• ML / LLM-based detection
• ...
Outputs: Potential vulnerabilities or suspicious locations
Detected Vulnerabilities
2
Vulnerability Analysis
Understand and assess vulnerability properties.
Typical Tasks
• Root-cause identification
• CWE / taxonomy mapping
• Severity assessment (CVSS, etc.)
• Impact analysis
• Exploitability assessment
• ATT&CK mapping
• ...
Outputs: Explained and prioritized vulnerabilities
Prioritized / Explained Vulnerabilities
3
Vulnerability Mitigation
Reduce or eliminate risks and prevent future exploitation.
Typical Tasks
• Repair recommendation
• Patch generation / code modification
• Patch validation / regression testing
• Functional preservation
• Mitigation guideline generation
• ...
Outputs: Patches, validated fixes, and mitigation guidance
Mitigation Artifacts
Patches
Repair Patterns
Tested Examples
Best Practices
...